

ENGINEERING WORKSHOP

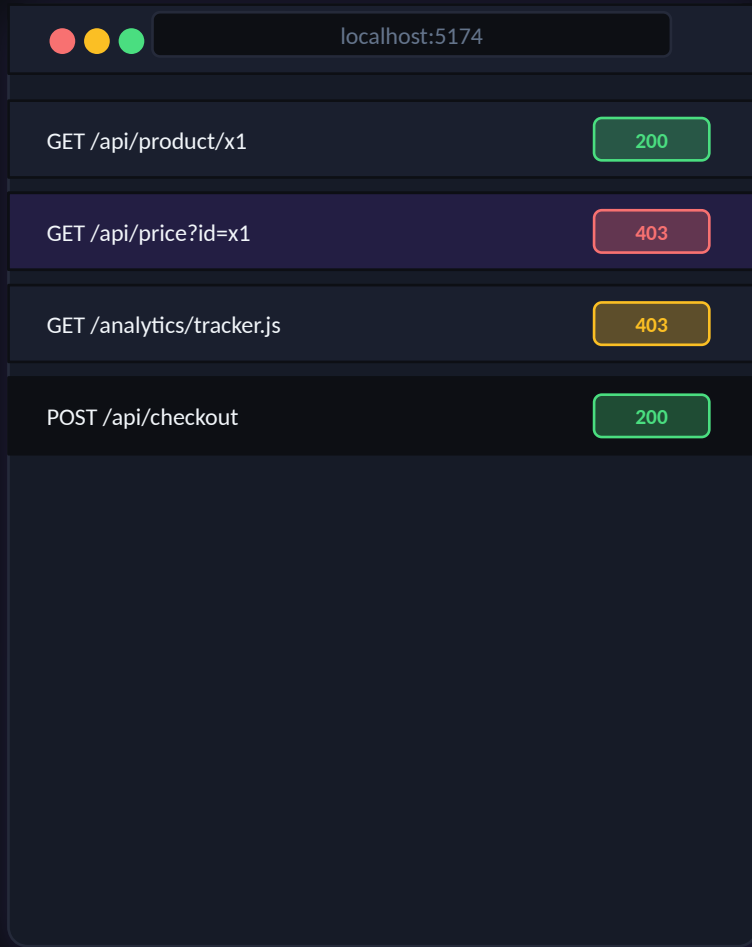
Chrome DevTools Playground

Debug production issues like a real engineer.

Frontend

Fullstack

QA



THE PROBLEM

Production debugging is different.

You can't reproduce most production bugs in development — the conditions don't exist locally.



Edit source code



Restart production



Change API responses



Disable 3rd-party services

*Instead... You **investigate**.*

Teach practical Chrome DevTools skills

through a realistic shopping experience.

 **Not tutorials**

 **Real debugging**

Every scenario mirrors a real production failure mode.

THE SETUP

A user wants to
buy headphones.



Wireless Pro Headphones X1

\$49

~~\$999~~

THE MISSION

Complete checkout using only
Chrome DevTools.

4 gates. 4 skills. 0 code changes.

Everything. Goes. Wrong.

NOT ALLOWED



Source code edits



Backend access



Restart the app

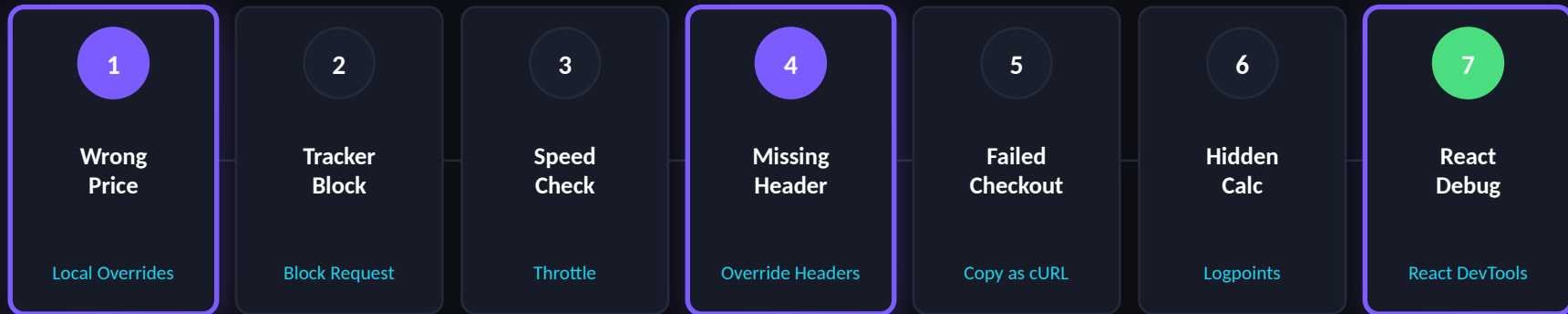
ONLY THIS



Chrome DevTools

Your only weapon.

7 scenarios. Each one a real skill.



SCENARIO 01

Wrong Price

PROBLEM

Product shows \$999 instead of \$49 due to a stale API response.

DEVTOOLS FEATURE

Local Overrides

REAL-WORLD EQUIVALENT

Backend bug

Data validation

WHAT YOU LEARN

Override any API response without touching source code.

DIFFICULTY



Response Preview — /api/price?id=x1

```
{  
  "id": "x1",  
  "name": "Headphones X1",  
  "price": 999,  
  "currency": "USD"  
}
```

OVERRIDE →

```
{  
  "id": "x1",  
  "name": "Headphones X1",  
  "price": 49,  
  "currency": "USD"  
}
```

SCENARIO 02

Tracker Block

PROBLEM

An analytics tracker is slowing the page and blocking checkout.

DEVTOOLS FEATURE

Block Request

REAL-WORLD EQUIVALENT

3rd-party scripts

Performance

WHAT YOU LEARN

Block specific network requests to isolate script-caused failures.

DIFFICULTY



Network — Request Waterfall

GET /api/product/x1



GET /analytics/tracker.js



BLOCKED

GET /api/checkout



SCENARIO 03

Speed Check

PROBLEM

Checkout is slow on mobile networks but fast on desktop.

DEVTOOLS FEATURE

Per-Request Throttle

REAL-WORLD EQUIVALENT

Mobile UX

Network conditions

WHAT YOU LEARN

Throttle specific requests to simulate real-world mobile latency.

DIFFICULTY



Network — Per-Request Throttle

GET /api/product/x1



12ms

GET /api/ping

THROTTLED 2.4s

Throttle Profile: 3G Mobile — 750 kb/s down 250 kb/s up 100ms RTT

Override Headers

PROBLEM

Feature flag requires a custom header you can't add in production.

DEVTOOLS FEATURE

Override Headers

REAL-WORLD EQUIVALENT

Feature flags

A/B testing

WHAT YOU LEARN

Inject custom HTTP headers into outgoing requests without code.

DIFFICULTY



Response Headers — /api/checkout

content-type: application/json

x-request-id: abc123

test-header-value: 1 ← INJECTED

cache-control: no-store

Failed Checkout

PROBLEM

Checkout API returns an error that only happens in this browser session.

DEVTOOLS FEATURE

[Copy as cURL](#)

REAL-WORLD EQUIVALENT

[API debugging](#)[Bug reproduction](#)

WHAT YOU LEARN

Reproduce any browser request exactly in terminal to isolate the issue.

DIFFICULTY



bash

```
$ curl -X POST https://app.localhost/api/checkout-  
broken \  
  -H 'Content-Type: application/json' \  
  -H 'test-header-value: 1' \  
  -d '{"productId":"x1","coupon":"SAVE20"}'  
  
{  
  "error": "invalid_coupon",  
  "message": "Coupon SAVE20 expired"  
}
```

SCENARIO 06

Hidden Calculation

PROBLEM

A discount calculation is producing wrong totals — no error thrown.

DEVTOOLS FEATURE

Logpoints

REAL-WORLD EQUIVALENT

Silent bugs

Business logic

WHAT YOU LEARN

Insert console log statements into running code without editing files.

DIFFICULTY



Sources — checkout.js

```
1  function applyDiscount(price, coupon) {  
2      const rate = COUPON_RATES[coupon];  
3      const discount = price * rate;  
4      return price - discount;  
5  }
```

Console

```
◆ price=49 rate=undefined discount=NaN  
◆ price=49 rate=0.2 discount=9.8
```

SCENARIO 07

React Debug

PROBLEM

The BuyButton component doesn't trigger despite state appearing correct.

DEVTOOLS FEATURE

React DevTools

REAL-WORLD EQUIVALENT

Component state

React props

WHAT YOU LEARN

Inspect and manipulate React component state in real time.

DIFFICULTY



React DevTools — Components

- ▶ App
 - ▶ ProductSection
 - ▶ BuyButton
 - ▶ Spinner

State

buying: false
order: null

Suspense

Error Boundary

Profiler

State Editor

Before → After

BEFORE

"I can't reproduce it."

File a ticket

Wait for logs

Hope it happens again



AFTER

*"I know exactly
why it happens."*

Override the response

Block the script

Throttle the request

TAKEAWAYS

By the end, you'll debug production like a senior engineer — without touching a single line of code.

- ✓ Debug production safely
- ✓ Validate backend assumptions
- ✓ Isolate third-party failures
- ✓ Simulate network conditions
- ✓ Investigate without code changes
- ✓ Produce actionable bug reports

“Think like a production engineer.”